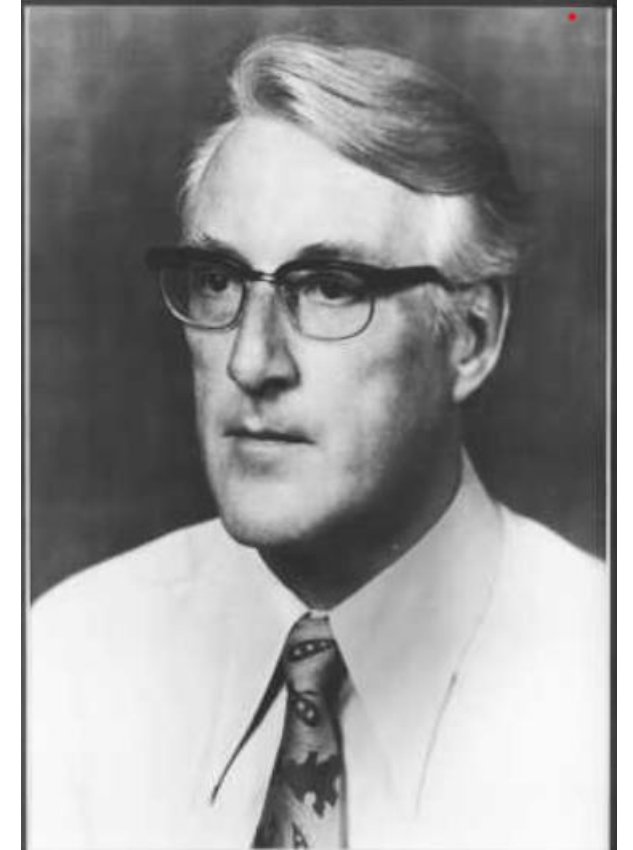


The early-career returnee's dilemma

E D Davis, Sol Plaatje University
(david.davis@spu.ac.za)

Wilhelm E. Frahn

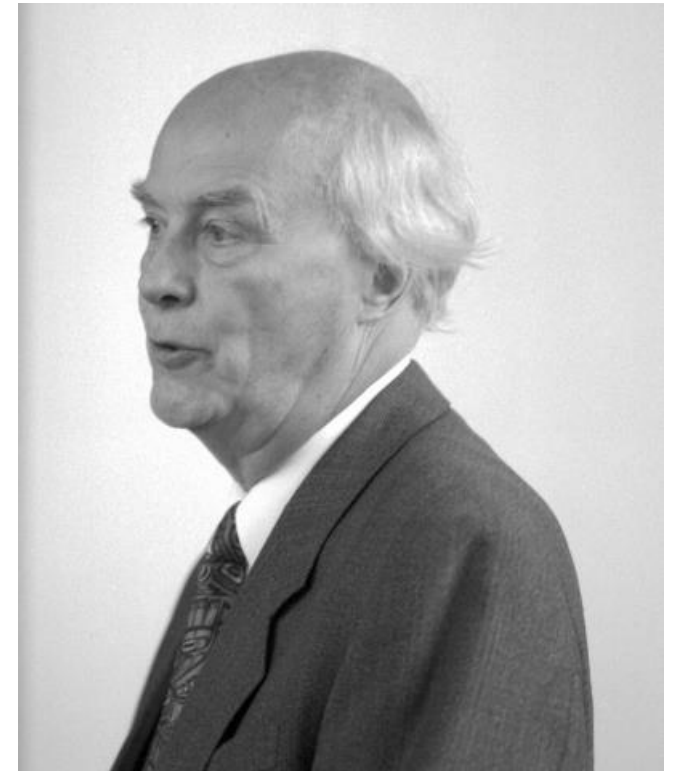
- South Africa: longstanding tradition of importing theorists from Germany
- First theorist in SA: “Willie” Frahn, appointed to CSIR in 1955
- Inspirational lectures on QM & quantum statistical mechanics
- “The art of the theorist is to identify a small parameter and expand in it”



W.E. Frahn (1926-1982),
Dr. rer. nat. (Aachen, 1953)

My debt to Germany

- Trained by WE Frahn (UCT) & WD Heiss (Wits)
- Offered my first post-Doc by Max-Planck-Institut f. Kernphysik, Heidelberg
- Fortunate to work in Hans Weidenmüller's group for 3 years – a *transformative* experience

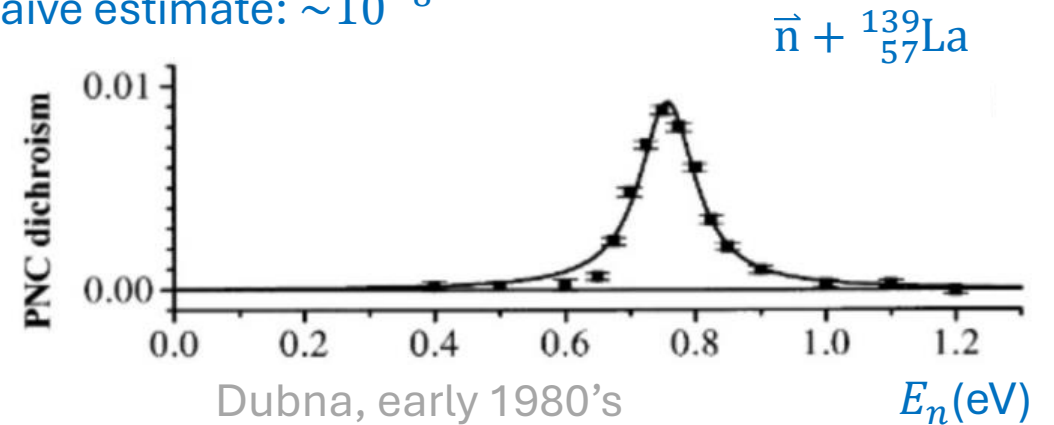


H.-A. Weidenmüller (1933-),
Dr. rer. nat. (Heidelberg, 1957)

TRIPLE@LANL

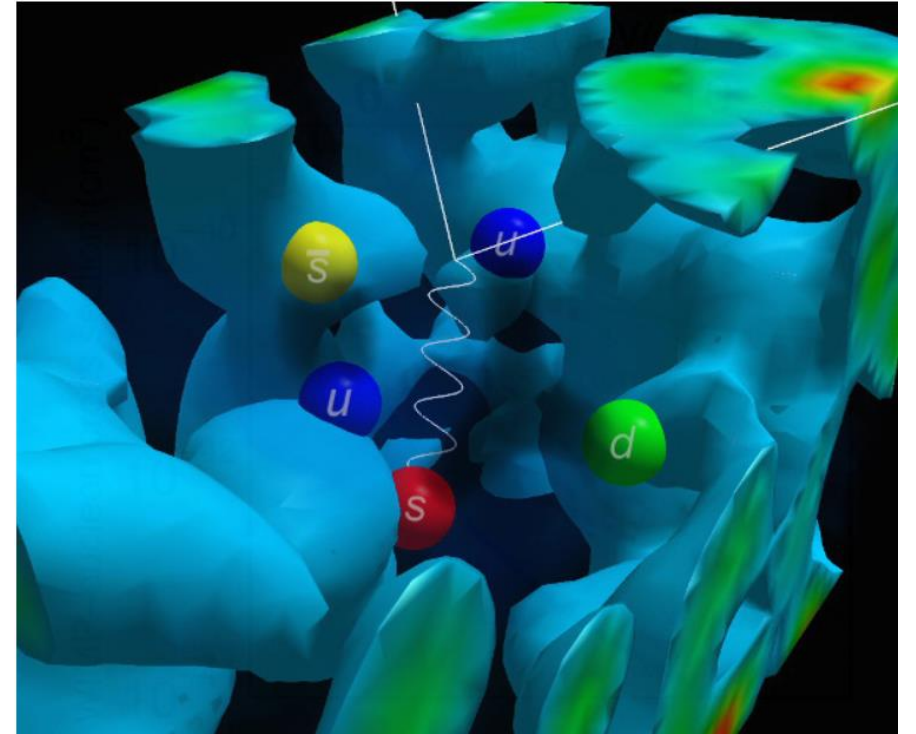
- Probe CP-violation up to the 100 TeV scale with epithermal neutrons (Beyond Std Model physics)
- Resonance enhancement by a factor of 10^6
- Random matrix analysis possible (ensembles determined by space-time symmetries)

Naïve estimate: $\sim 10^{-8}$



A continuous thread

- Low-energy high-intensity (nuclear) precision probes of BSM physics
- Need better nuclear matrix elements
- Quantum computing *may* help



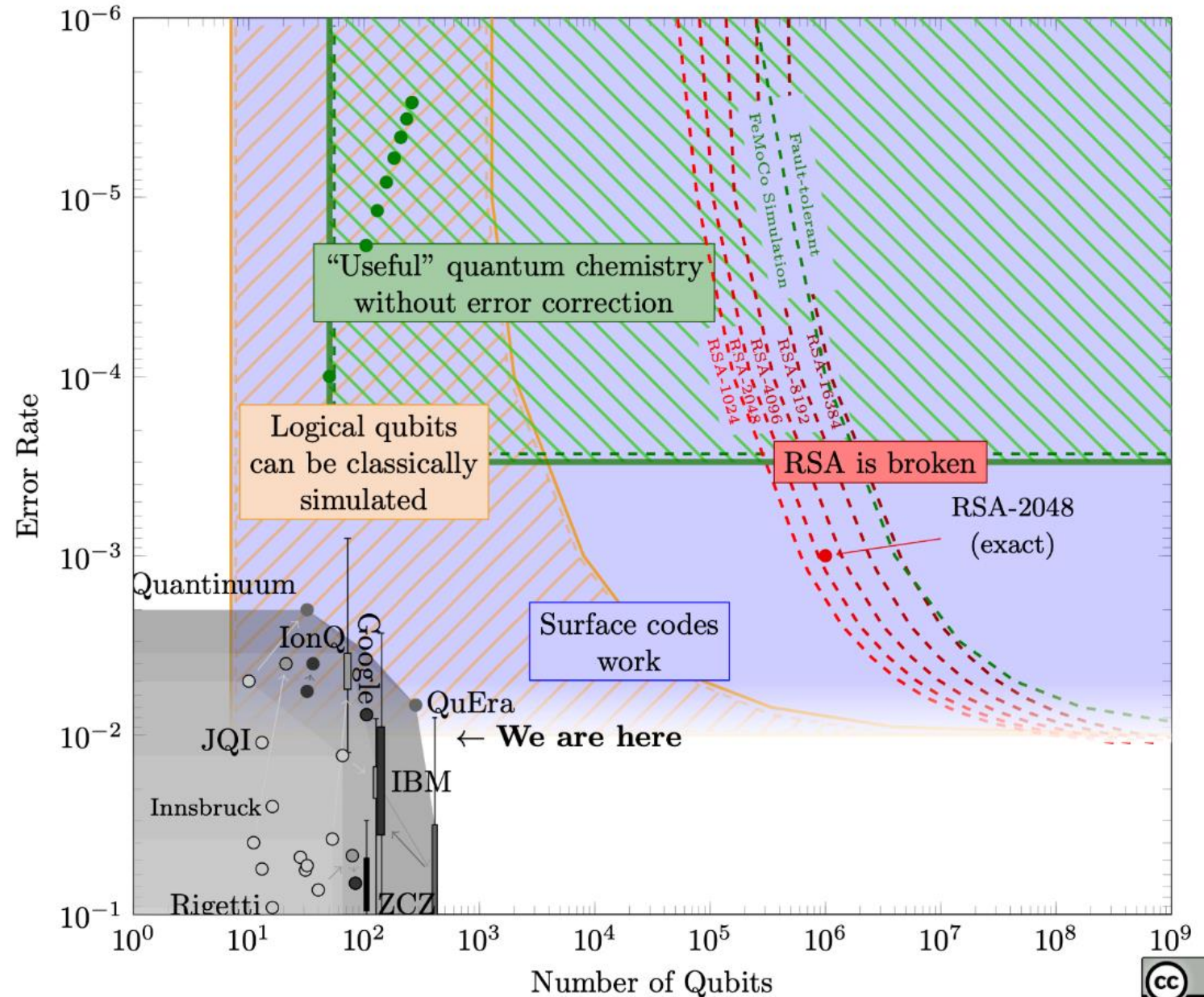
Derek Leinweber, 2017

QC Baedeker

Samuel Jacques
(U. Waterloo)

On breaking RSA encryption: “it’s a race between quantum engineers and system administrators, where the system administrators have a 30 year headstart.”

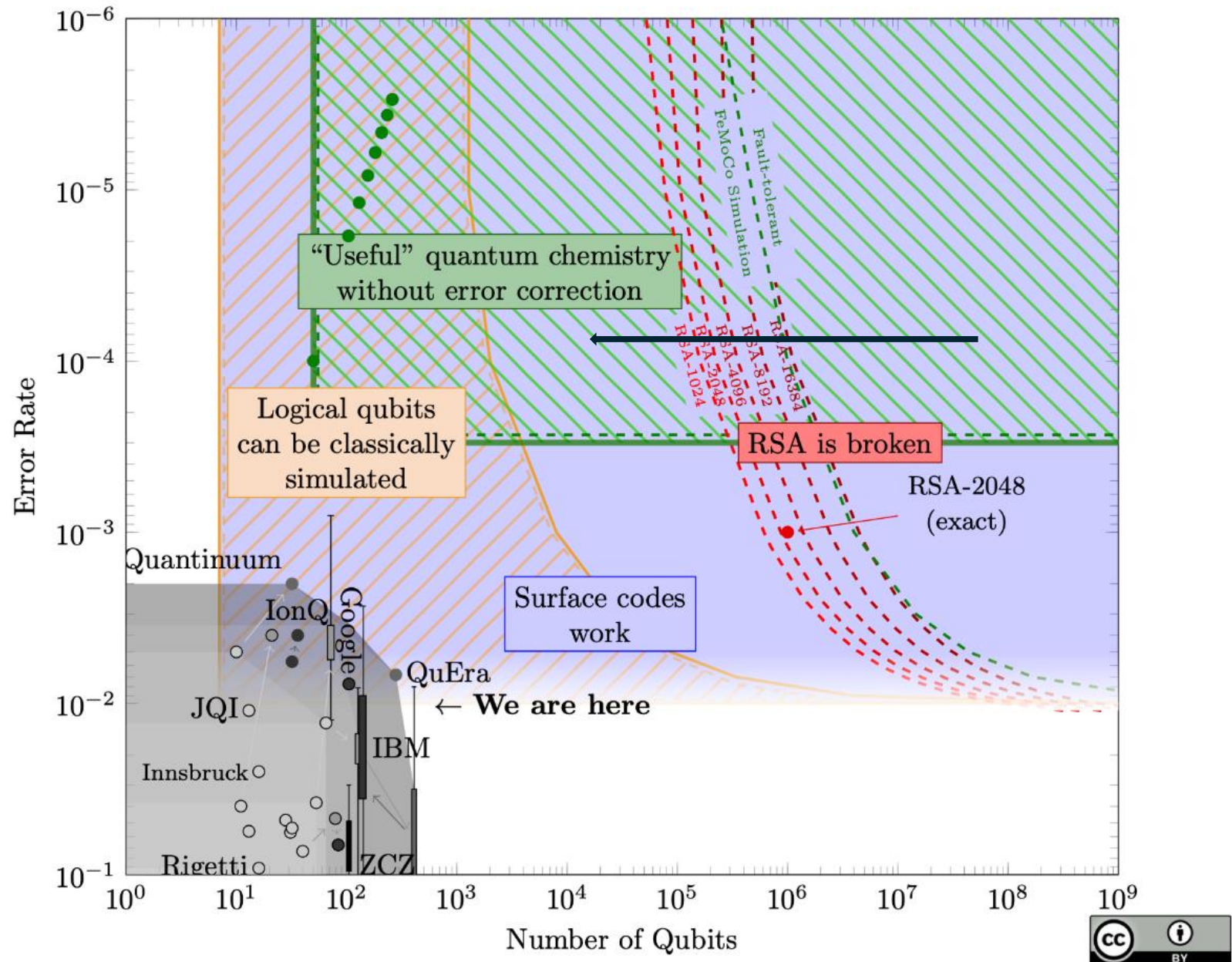
Landscape of Quantum Computing in 2025



My interest

Hilbert space is a big place.

Can physically-motivated schemes enhance the features of interest?

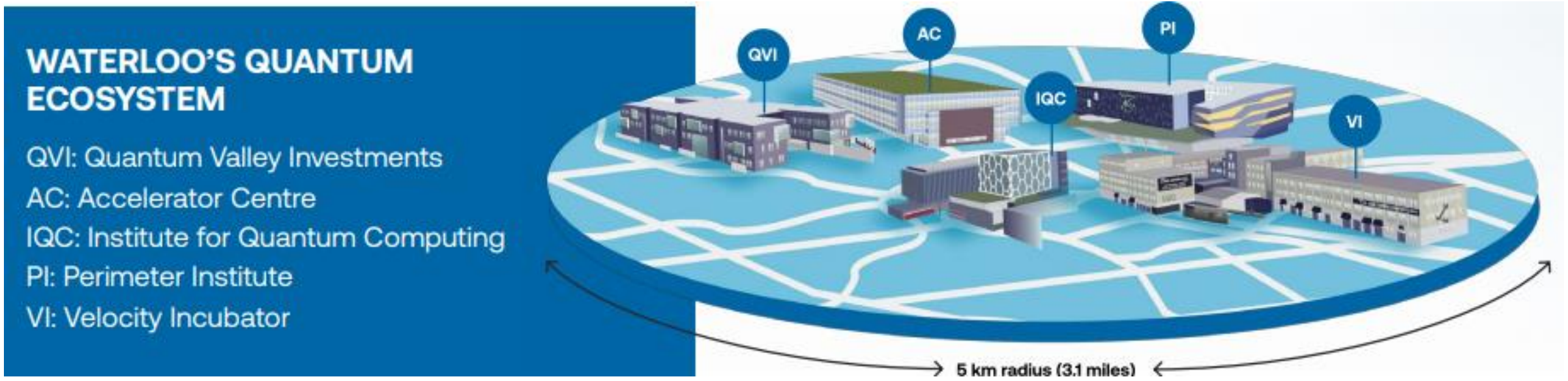


Some of the issues faced by early-career returnees

- **Substantial teaching load**
- Counter-productive research output demands
- Infrequent visitors/Very limited international exposure
- Nepotism/Inbreeding
- Antipathy towards fundamental research

A solution? “Quantum Valley”

A new supra-national facility modeled on Waterloo’s Quantum Ecosystem



Conclusion

- Theoretical physics training in South Africa has been profoundly influenced by the German tradition
- Nevertheless, post-Doctoral experience abroad of several years is an essential ingredient
- We **need** to establish something like a Quantum Valley to take **full** advantage of the talents and potential of returning post-Docs